

Nitya Lagadapati (She/Her)
(425) 499-1791 • Nityalagadapati@gmail.com

EDUCATION

Johns Hopkins University, Baltimore, MD

Master of Science in Engineering (M.S.E), Biomedical Engineering August 2025-May 2026

Translational Cell and Tissue Engineering, and AI in Medicine Tracks; GPA: 3.94

Relevant Coursework: Precision Care Medicine, Cellular Engineering, Tissue Engineering, Translating Healthcare AI: Essential Concepts, Cell and Tissue Engineering Lab, and Immunomodulatory Biomaterials

University of Washington, Seattle, WA

Bachelor of Science, Bioengineering: Data Science; GPA: 3.89

September 2021-June 2025

Informatics Minor

Relevant Coursework: Biomolecular Engineering, Biomedical Signals and Sensors, Biotransport I-II, Solid & Gel Mechanics, Systems and Controls, Fundamentals of Bioengineering Design, Failure Analysis & Human Physiology, Tissue Engineering, Systems Immunology, Quantitative Physiology, Biosensors for Cancer, Computational Modeling & Simulation of Bioelectricity, Nanotechnology, and Optogenetics

HONORS

- *Cum laude*, University of Washington College of Engineering
- Mary Gates Research Scholar (Winter 2025), University of Washington
- Hershel D. Graves Endowed Scholarship in Engineering (2024-2025), University of Washington Department of Bioengineering
- Annual Dean's List (2021-2025), University of Washington

RESEARCH EXPERIENCE- ACADEMIA

Wilson Lab, Johns Hopkins University

Graduate Researcher, March 2026 – Present

PI: Dr. Scott Wilson

- Developing a polymer-antigen conjugate-based influenza vaccine platform that targets antigen-presenting cells to enhance immune response efficacy.
- Utilizing spin column chromatography, HPLC, and SDS-PAGE to purify, characterize, and validate conjugation efficiency and product quality.

Kathuria Lab, Johns Hopkins University

Graduate Researcher, December 2025 – Present

PI: Dr. Annie Kathuria

- Analyzing electrophysiology datasets from stem cell-derived organoid models of neurological pathologies, to characterize responses to pharmacological therapeutic interventions.
- Developed a Python pipeline to automate extraction and organization of electrophysiology reports.
- Generating publication-quality methods and data schematics using Origin and BioRender.

Tissue and Regenerative Engineering Lab, University of Washington

Undergraduate Researcher, October 2023 – July 2025

PI: Dr. Jenny Robinson, Mentor: Kiley Burkey (MD/PhD student)

Project: Tuning Viscoelasticity in a Hyaluronic Acid-Based Hydrogel System to Investigate Meniscal Cell Mechanotransduction.

- Developed a hyaluronic acid-based hydrogel system that is able to mimic biomechanical properties observed in the native meniscus.
- Performed and optimized swelling studies and uniaxial compression and stress relaxation tests on a dynamic mechanical analysis system.

- Developed MATLAB code that can process the data returned from the testing equipment into interpretable values and test different analysis methods.
- Trained an undergraduate student in hydrogel fabrication and operation of testing equipment.

Stevens Lab, University of Washington

Undergraduate Researcher, September 2022 – June 2023

PI: Dr. Kelly Stevens, Mentor: Susana Simmonds (PhD student)

Project: 3D-Printed Hydrogel Microarray for High-Throughput In Vivo Biomaterial Screening in Large Rodents.

- Designed a biomaterial-based microarray that would be implanted in large rodents to perform high-throughput designer tissue screening for liver tissue engineering applications using SolidWorks and 3D printed multiple iterations.
- Modulated designs to include microwells of different volumes and shapes to investigate impact on cellular adhesion and behavior.
- Learned and practiced cell culture techniques and created an experimental plan to test the efficacy of different microarray designs in preparation for in-vitro tests.

RESEARCH EXPERIENCE- INDUSTRY

Partner Therapeutics, Lynnwood, WA

Downstream Purification Intern, June 2024 – September 2024

- Performed high-throughput screening of drug product purification solvents using 96-well plate chromatography resins with the goal of changing organic chromatography solvents to aqueous solvents.
- Narrowed down successful conditions by repeating studies at the larger scale and recommended conditions equivalent to current product purification standards for further studies.
- Screened second source resins for the commercial process through HPLC (semi-prep and analytical scale) experiments.
- Gave a cumulative presentation to the Process Science group at the end of the internship and created a poster presentation for an innovation symposium within the company.

Lundbeck Seattle Biopharmaceuticals, Bothell, WA

Formulation Development and Drug Product Intern, June 2023 – September 2023

- Designed and executed experiments investigating the effects of buffer pH, additives, and temperature on product viscosity and protein-protein interactions.
- Analyzed experimental data using JMP software to assess formulation efficacy and identify the relative impact of key variables.
- Utilized industry-standard tools and adhered to general manufacturing practices.
- Delivered a 20-minute presentation summarizing motive and findings and offering actionable insights for optimizing product formulations.

PROJECTS

LLM-Driven Clinical Information Extraction for Immunotherapy Modeling

Precision Care Medicine Class, JHU, September 2025 – Present

- Developing and optimizing prompt engineering techniques to improve structured data extraction from radiographical reports for the development of downstream predictive models for immunotherapy resistance.

Blink Detection Algorithm

Independent Project: Center for Language and Speech Processing, JHU, September 2025 – Present

- Developing and refining a blink detection algorithm to support a clinical tool aimed at early

- diagnosis of neurodegenerative diseases.
- Contributing to the transition of data processing pipelines between different eye-tracking devices.

Synthesized Breath Sounds

Capstone Project: Department of Bioengineering & CREST Lab, UW, January 2025 – June 2025

- Optimized generative machine learning models to synthesize realistic breath sounds tailored to specific diagnoses, patient demographics, and stethoscope placement.
- Experimented with data augmentation techniques to expand the training and testing data.
- Aimed to integrate the synthesized breath sounds into training manikins to improve clinical education and diagnostic skills.

Ultrasound Training Phantom for Jugular Vein Injections

Term Project: Fluid and Materials Lab, October 2023 - December 2023

- Developed a ballistic gel model of the neck with active fluid flow to simulate physiological conditions.
- Designed a system compatible with a linear array ultrasound probe to help healthcare professionals practice and improve accuracy in performing jugular vein injections.

Endozone, Bioengineers Without Borders Subproject

Member, September 2022-October 2023

- Conceptualized and prototyped a non-invasive process for diagnosing endometriosis using a lateral flow assay.
- Literature analysis of biomarkers associated with endometriosis and their identification process.
- Raised awareness of endometriosis by creating demonstrations and materials for use in outreach events.

CGeneIE: Cancer Gene Identification Environment

Term Project: Intermediate Data Programming, May 2023- June 2023

- Experimented with Naïve Bayes, Random Forest, and Test Ridge classification-based machine learning algorithms and their parameters in Python
- Assessed their accuracy in predicting types of breast cancers using gene sequence and protein sequence data from breast cancer patients.

PRESENTATIONS

“Tuning Viscoelasticity in a Hyaluronic Acid-Based Hydrogel System to Investigate Meniscal Cell Mechanotransduction.” **Lagadapati, N.**, Burkey, K., & Robinson, J.

- Poster Presentation at Society for Biomaterials Annual Meeting and Exposition, April 2025
- Rapid-Fire and Poster Presentation at Society for Biomaterials- Northwest Regional Conference, September 2024
- Poster Presentation at Undergraduate Research Symposium, University of Washington, May 2024
- Poster Presentation at Stem Cell Symposium, Institute of Stem Cell and Regenerative Medicine at the University of Washington, May 2024

“Development of an Aqueous Alternative to the C4 Capture Step” **Lagadapati, N.** Oral Presentation at Partner Therapeutics Process Science Meeting, September 2024

- Poster presentation at the Partner Therapeutics internal Innovation Symposium, October 2024

“Optimizing [Redacted] Formulations: Impact of pH, Additives, and Temperature on Viscosity and Protein Interactions.” **Lagadapati, N.** Oral Presentation at Lundbeck Seattle Biopharmaceuticals Sitewide Meeting, September 2023

LEADERSHIP/JOB EXPERIENCE

STEM Achievement in Baltimore Elementary Schools, Baltimore, MD

Afterschool STEM Teacher, September 2025 – Present

- Facilitating hands-on inquiry-based STEM learning for 4-5th grade students using the Engineering Adventures curriculum.
- Creating engaging lesson plans to guide students through engineering design challenges.
- Collaborating with STEM mentors and professionals to support student engagement, integrating the Engineering Design Process to promote critical thinking, problem-solving, and iterative design improvement.

Bioexplore, Seattle, WA

Mentor, November 2024 – June 2025

- Served as a mentor to two freshmen in the College of Engineering at the University of Washington.
- Provided advice on coursework, navigating college, finding research opportunities, as well as career paths/academic trajectories.

Bioengineers Without Borders, Seattle, WA

Co-President, September 2023 – June 2025

Vice President of External Communications, February 2022- September 2023

- Registered student organization at the University of Washington aiming to coordinate collaboration between students from the fields of engineering, physics, biology, mathematics, and related sciences to develop low-cost, sustainable medical technologies through independent research, design, and development projects.
- Coordinated between 4 subprojects and external collaboration and funding opportunities.
- Supervised the rest of the board in managing social media, advertising events, and managing funds.
- Ideated outreach events and future projects and execute recruitment events.
- Organized supplies and led K-12 students in understanding the engineering design process while developing a water filtration system during UW's Discovery Days event.

University of Washington Department of Bioengineering, Seattle, WA

Teaching Assistant, September 2023 – December 2023

Course: MATLAB Fundamentals for Bioengineers

- Formulated and taught biweekly lectures on a variety of topics including image processing, modeling biological processes using ordinary differential equations, statistical analysis and representations, and biological simulations (ex: spread of a virus).
- Incorporated engaging activities into lectures and assignments to meet the needs of students.
- Hosted weekly office hours, supported students in completing weekly assignments and a final project, and gave constructive feedback during the grading process.

TECHNICAL SKILLS

Computational: Python, MATLAB, R, SQL, Java, JMP, GraphPad Prism, SolidWorks, AutoCAD, Machine learning applications, Microsoft Office Suite, Google Suite

Wet Laboratory: Mechanical analysis testing using a dynamic mechanical analysis system, cell culture, buffer formulation, HPLC (high-performance liquid chromatography) operation, UPLC (ultra-high-performance liquid chromatography) operation, protein purification techniques, and UNCLE (protein stability screening platform) operation.

Languages:

Telugu: Native proficiency, French: Limited working proficiency